ISHAQ PAKTIN YAR

Data Scientist · Backend Engineer · ML Researcher

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ML Engineer with a track record of delivering production-grade systems and measurable research impact; **18% forecasting lift, 98% protein classification accuracy, 60% reduction in admin overhead.** Unique combination of deep learning research (CU Anschutz), full-stack production engineering, and academic excellence (B.Sc. High Honors, 2025). Fluent in 5 languages across 4 language families.

EDUCATION

B.Sc. Computer Engineering — High Honors · *Istanbul Arel University* 2021 – 2025

Thesis: *ClimatePulse: Multivariate Time-Series Forecasting for Air Quality & Climate* — Achieved 18% MAE improvement over ARIMA and LSTM baselines using multi-horizon deep learning (NeuralForecast + PyTorch).

Supervisor: Asst. Prof. Mert Yağcıoğlu

WORK EXPERIENCE

Undergraduate Researcher · *JRAVI Lab — CU Anschutz · Remote* Jul 2024 – Jun 2026

- Improved protein function prediction accuracy from 86% to 98%
- Benchmarked lightweight models against protein language models at significantly lower compute cost
- Contributing to manuscript in preparation for peer-reviewed submission.

Software Engineer · *Parem Academy · Istanbul* Aug 2025 – Aug 2026

- Architected 5+ production systems including Bitatlım and Veroz serving real users
- Designed high-concurrency REST APIs and real-time tracking pipelines using Node.js and PostgreSQL
- Led infrastructure decisions cutting hosting costs by 90%.

IT Coordinator · *Istanbul&I · Istanbul* Sep 2023 – Sep 2025

- Designed and deployed Django-based system serving 1,800+ users, replacing fragmented Excel workflows
- Built secure REST APIs for member analytics and operational tracking
- Reduced administrative processing time by 60%

Software Developer Intern · *DevSecure · Istanbul* Feb 2025 – Aug 2025

- Maintained and debugged client web platforms within active production environments
- Mentored 2–3 junior interns on Git workflows and deployment standards

Science & Programming Educator · *Bilim Üsküdar Science Center · Istanbul* Dec 2023 – Apr 2024

- Taught applied science and programming to 20+ students daily
- Designed original lesson demystifying neural networks for young learners

KEY PROJECTS

Bitatlım — Hyperlocal Marketplace & Logistics Platform *Next.js · NestJS · PostgreSQL · Redis · Linux VPS*

- Architected multi-tenant marketplace with hybrid logistics engine supporting real-time courier pooling
- Reduced hosting costs by 90% via optimized VPS infrastructure and automated CI/CD

Pathogenic Feature Discovery — Deep Sequence Modeling *Python · PyTorch · SHAP*

- Built Conv1D + Attention model for pathogenicity prediction from raw protein sequences
- Applied SHAP-based auditing to validate motif learning and rule out taxonomic bias

Air Quality Multivariate Forecasting (Thesis) *Python · PyTorch · NeuralForecast*

- Developed 7-day multi-horizon forecasts for PM10, NO2, SO2, O3 achieving 18% MAE improvement over ARIMA and LSTM baselines
- Engineered a RASA NLU chatbot enabling natural-language weather queries with real-time forecast integration

Additional projects available at Portfolio

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, Bash, Java, C++
ML / AI	PyTorch, Transformers, CNNs · RNNs · LSTMs, Time-Series Forecasting, SHAP
Backend	Node.js (NestJS / Express), Django, Flask, REST APIs, ETL Pipelines
Data	PostgreSQL, Redis, Relational Modeling, Real-Time Pipelines
Infrastructure	Docker, Linux, AWS, Git, CI/CD, Self-Hosted VPS

LANGUAGES

Pashto (Native) **English** C1 **Persian** C1 **Urdu** C1 **Turkish** B2 **Spanish** A2

CERTIFICATIONS & COMMUNITY

IBM AI Engineering — *IBM Coursera* (2023) · **IELTS** — *British Council* (2024)
Founder, *Istanbul&i Book Club* (2024) · **Data Commission Volunteer**, *T3 AI'LE* (2024)
Local Team Coordinator, *Best Diplomats* (2024) · **Volunteer**, *Sokak Lambası Association* (2023–Present)